A LIVE LINUX DISTRIBUTION FOR TACTICAL INFORMATION AND EMERGENCY COMMUNICATION

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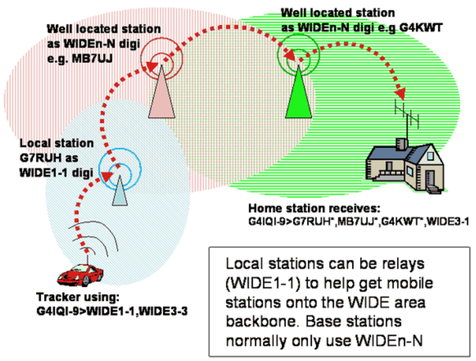
Summary

Amateur radio (also called ham radio) operators play a vital role in providing communication service in emergency situations such as earthquake, fire, and hurricane. Those operators provide different types of communication including voice on different frequencies and data with geographic information using Automatic Packet Reporting System (APRS).

In this work, a live Linux distribution coupled with APRS-aware software that can be used from a USB stick is presented. This distribution focuses on tactical digital communication using APRS and presents a pre-defined configuration of the software before an emergency. Such configuration includes offline map data and set of tools that are useful in an emergency within a USB stick. This USB stick can play a vital role for providing tactical communication during an emergency when there is no Internet connectivity.

General Overview of APRS



A mobile station has radio with APRS capabilities. This station receives location from GPS and sends this information to APRS network using a radio. A repeater station hears the message and repeats it, then another station repeats the same message. In the end, a home station that is far away from the mobile station gets the location information of the mobile station. What this implies is that even if Internet, phone exchanges, 3G and other communication systems are lost in the emergency, it is still possible to communicate and get tactical information using radios.

Motivation

APRS is used by interfacing a radio (hand-held or stationary) with a computer via serial port. The APRS-aware software reads data from radio and shows the related items in the map. To achieve this, a set of software should be present in the computer (operating system, software, its configuration, and offline map data). However, it is not possible for a ham radio operator to install this configuration to every laptop present. In an emergency, it is very probable to forget the laptop which has APRS software installed. Nevertheless, it is much easier to find an arbitrary laptop in Disaster Control Centers, which can be used for communication purposes.

This USB stick is easy to carry while holding offline map and a software (Xastir) to be interfaced with a radio. When disaster happens, it will be enough for ham radio operator to run the USB stick on an arbitrary computer, connect the radio, and start getting tactical information. Effectively, this distribution turns any computer into tactical communication center.

Screenshots From The Running System

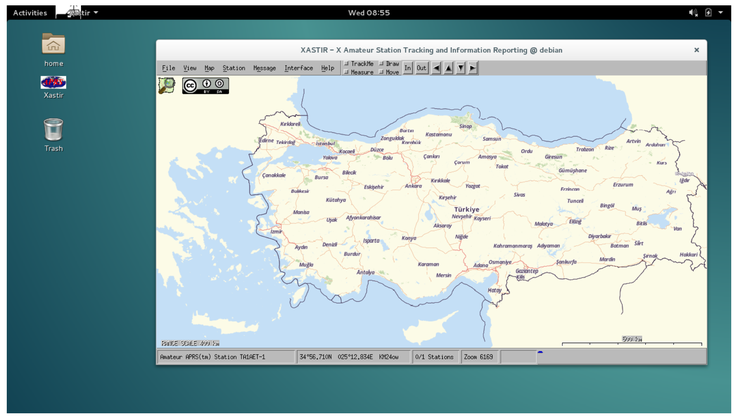


Figure: Main desktop with Xastir running using OpenStreetMap (OSM) data.

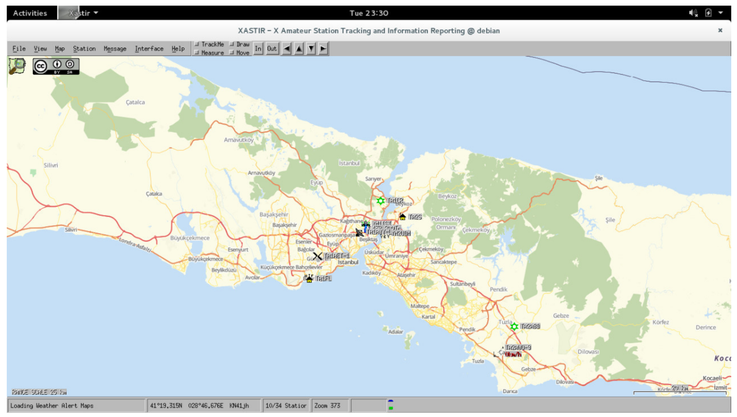


Figure: Stations displayed on the map with zoom.

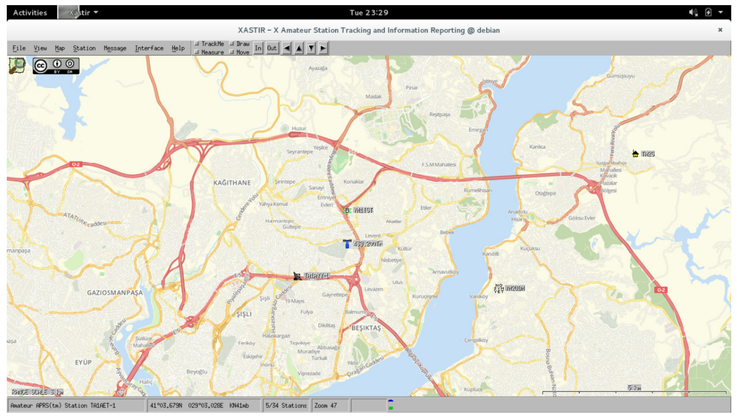


Figure: Another zoom level. It is possible to zoom up to the street level for Istanbul. This map data is pre-rendered and the stations shown here are received using only a radio. No other connectivity is present.

Creating Live Distribution And Setting It Up

Debian live build project [1] *(version 4.0.3.-1)* is used to create the live USB disk. The resulting operating system is based on Debian Jessie (8.0). The resulting binary image is written to the usb disk under Ubuntu 14.04 with *dd if=live-build.iso of=/dev/sdX* command where X is the usb device. Live build downloads large amount of packages from debian repositories to create an image. This leads to longer build time and configure-make-build-test cycle. To reduce the amount of time spending on downloading the packages, *apt-cacher-ng* is used to cache packages, which improved the build time significantly.

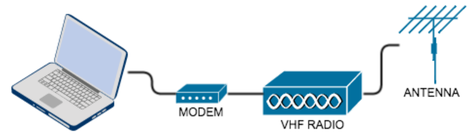


Figure: Connection diagram of the components for getting tactical information.

The image comes with Xastir version *2.0.4-2+b1*. Xastir is an APRS client for Linux and it provides *mapping, tracking, messaging, weather, weather alerts, and Search/Rescue features over radio* [2]. APRS data is received using Argent Data OpenTracker+ [3] connected to Very High Frequency (VHF) radio. Keep It Simple, Stupid (KISS) protocol firmware *(version s19)* is used in the tracker. OpenTracker+ acts as a modem between the radio and the computer. Its main function is to demodulate APRS signal received from radio and send digital packets via serial port using KISS [4] protocol. Modern computers, however, do not have a serial port so a serial-to-usb converter is used to interface OpenTracker+ with a computer. Xastir is configured to use KISS modem (Opentracker+) as a data source. Xastir initializes the modem, listens for the incoming packets on serial port, renders the items, and shows them in the map. With Xastir set up, an operator is able to get real-time tactical information, mark and send location as well as private messages to other operators. It should be noted that no other connectivity other than radio is present in this configuration.

Conclusion

The available Linux distributions aim to be generic and complete toolkit for amateur radio operators. There is little or no work on tactical information, Geographic Information System (GIS), or APRS specifically. Although APRS software is included along with other ham radio applications, the main aim of many hamradio distributions is not tactical information but rather being a general toolkit.

This work is aimed to make tactical communication easier by bundling APRS, GIS, offline map, and other tools with usability and emergency communication in mind. Effective communication is not possible without location information. This work can have an important role in providing tactical information for effective communication and the way we think about emergency situations.

References

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